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Equilibrium Performance Analysis of a LOX/LCH$_4$ Rocket Engine Using Cantera

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Abstract

This project presents a Python/Cantera analysis of an idealized methane-oxygen rocket combustion chamber. The study focuses on the influence of oxidizer-to-fuel mass ratio and chamber pressure on equilibrium combustion temperature, gas properties, product composition and ideal nozzle performance. The chemical state is obtained by adiabatic constant-pressure equilibrium calculations with Cantera and the high-temperature GRI-Mech mechanism. The nozzle calculation uses a frozen-gamma ideal-gas approximation to estimate characteristic velocity, thrust coefficient and specific impulse for a fixed nozzle area ratio. The results show that the maximum adiabatic flame temperature occurs close to the stoichiometric region, while the highest ideal vacuum specific impulse in this simplified model is shifted toward a fuel-rich mixture because molecular weight and characteristic velocity improve at lower O/F.

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1 Introduction

Liquid oxygen and liquid methane are an attractive propellant combination for modern rocket engines because methane is cleaner than kerosene, easier to store than hydrogen and compatible with reusability-oriented engine concepts. The pair is also relevant for in-situ resource utilization scenarios, where methane and oxygen can be produced from carbon dioxide and water. These practical advantages explain why LOX/LCH_4 engines are often considered in contemporary launch vehicle and exploration studies [1].

The objective of this project is to create a reproducible computational tool for a preliminary combustion analysis of a LOX/LCH_4 rocket chamber. The work is not intended to replace NASA CEA or a complete engine cycle model. Instead, it connects chemical equilibrium calculations with basic rocket-performance relations, producing engineering trends useful in early design reasoning.

The project deliverables are:

- Python source code based on Cantera,
- generated CSV result tables and figures,
- this LaTeX technical report,
- a small test suite checking the most important numerical helpers.

2 State of the Art

Chemical equilibrium is a classical first step in rocket propulsion analysis. NASA CEA calculates equilibrium compositions and theoretical rocket performance by minimizing thermodynamic potentials subject to elemental conservation [2]. The same type of equilibrium problem can be solved with Cantera, an open-source toolkit for chemical kinetics, thermodynamics and transport properties [3]. The present work follows that approach: Cantera supplies thermodynamic states and transport properties, while simple rocket equations are implemented directly in Python.

For rocket design, equilibrium calculations are useful because the chamber temperature, product molecular weight and heat capacity ratio strongly affect the characteristic velocity and nozzle expansion process [4]. However, the result must be interpreted as an ideal upper-bound trend. Real engines include incomplete mixing, finite-rate chemistry, pressure losses, wall heat transfer, two-phase effects when condensed species are present, boundary layers and efficiency losses in the nozzle.

3 Model Description

3.1 Reactant Composition

The reactants are methane and oxygen. For a specified oxidizer-to-fuel mass ratio, O/F, the corresponding mole numbers are calculated from molecular weights:

$$n_f = \frac{m_f}{M_f}, \qquad n_o = \frac{m_o}{M_o}, \qquad \frac{m_o}{m_f} = \mathrm{O/F}. \tag{1}$$

Cantera receives the inlet state as mole fractions:

$$X_f = \frac{n_f}{n_f + n_o}, \qquad X_o = \frac{n_o}{n_f + n_o}. \tag{2}$$

The default inlet temperature is 300 K. The baseline chamber pressure is 20 bar. The O/F sweep covers values from 2.0 to 4.4, including fuel-rich, nearly stoichiometric and oxidizer-rich conditions. For reference, the methane-oxygen stoichiometric O/F is approximately 4 when rounded to simple integer molecular weights.

3.2 Equilibrium Chamber State

The chamber is modeled as an adiabatic constant-pressure reactor. Cantera solves the HP equilibrium problem:

$$h(T, p, \mathbf{Y}) = h_0, \qquad p = p_c, \tag{3}$$

with elemental conservation enforced internally by the equilibrium solver. The gas mechanism is `gri30_highT.yaml`, which is suitable for the methane and oxygen gas-phase chemistry considered here. After equilibrium, the following properties are extracted:

$$\gamma = \frac{c_p}{c_v}, \qquad R = \frac{R_u}{\overline{M}}, \qquad a = \sqrt{\gamma R T_c}. \tag{4}$$

The dynamic viscosity, thermal conductivity and Prandtl number are also recorded from the Cantera transport model:

$$\mathrm{Pr} = \frac{c_p \mu}{k}. \tag{5}$$

3.3 Ideal Nozzle Model

The nozzle calculation is intentionally simple. The chamber equilibrium state provides T_c, γ and R. These values are then frozen and used in calorically perfect gas relations. The ideal characteristic velocity is

$$c^* = \sqrt{\frac{RT_c}{\gamma}} \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}. \tag{6}$$

For a fixed expansion area ratio $\varepsilon = A_e/A_t = 15$, the exit Mach number is found from the isentropic area relation:

$$\varepsilon = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}. \tag{7}$$

The exit pressure follows from

$$\frac{p_e}{p_c} = \left(1+\frac{\gamma-1}{2}M_e^2\right)^{-\gamma/(\gamma-1)}. \tag{8}$$

The thrust coefficient is calculated as

$$C_F = \sqrt{\frac{2\gamma^2}{\gamma-1}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}\left[1-\left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right]} + \frac{(p_e-p_a)A_e}{p_cA_t}. \tag{9}$$

Finally, the specific impulse is

$$I_{sp} = \frac{c^*C_F}{g_0}. \tag{10}$$

Both vacuum and sea-level values are reported. The sea-level case assumes $p_a = 1$ bar; the vacuum case uses $p_a = 0$.

4 Program Description

The code is split into a reusable model module and an analysis script:

- `src/rocket_combustion/model.py` contains the Cantera chamber calculation, O/F conversion, nozzle equations and sweep helpers.
- `scripts/run_analysis.py` performs the O/F and pressure sweeps, writes CSV files and generates figures.
- `tests/test_model.py` verifies the O/F conversion, area-Mach inversion and basic physical ranges of a representative case.

The main analysis can be reproduced with:

```
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe scripts\run_analysis.py
.\.venv\Scripts\python.exe -m unittest discover -s tests
```

5 Results

5.1 O/F Sweep

Figure 1 shows the adiabatic equilibrium chamber temperature. The curve rises rapidly from fuel-rich conditions and reaches a maximum of 3455.9 K at O/F = 3.70. The maximum is slightly fuel-rich relative to the rounded stoichiometric value O/F = 4.0 because dissociation and the detailed thermodynamic database shift the optimum temperature away from the simple global reaction estimate.

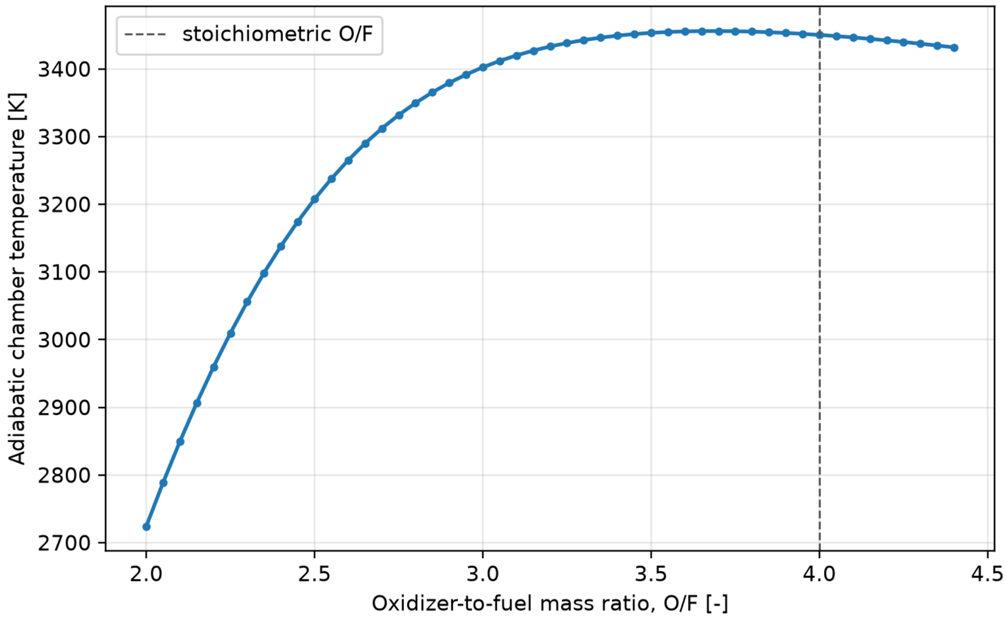

Figure 1: Adiabatic equilibrium chamber temperature as a function of O/F.

The performance trend in Figure 2 is different from the temperature trend. The maximum ideal vacuum specific impulse is 336.8 s at O/F = 2.55, while the maximum characteristic velocity is 1857.8 m/s at O/F = 2.50. This shift toward fuel-rich operation occurs because the products contain more H_2 and CO, which lowers the mean molecular weight. In the simplified frozen-gamma nozzle model, lower molecular weight can compensate for a lower combustion temperature.

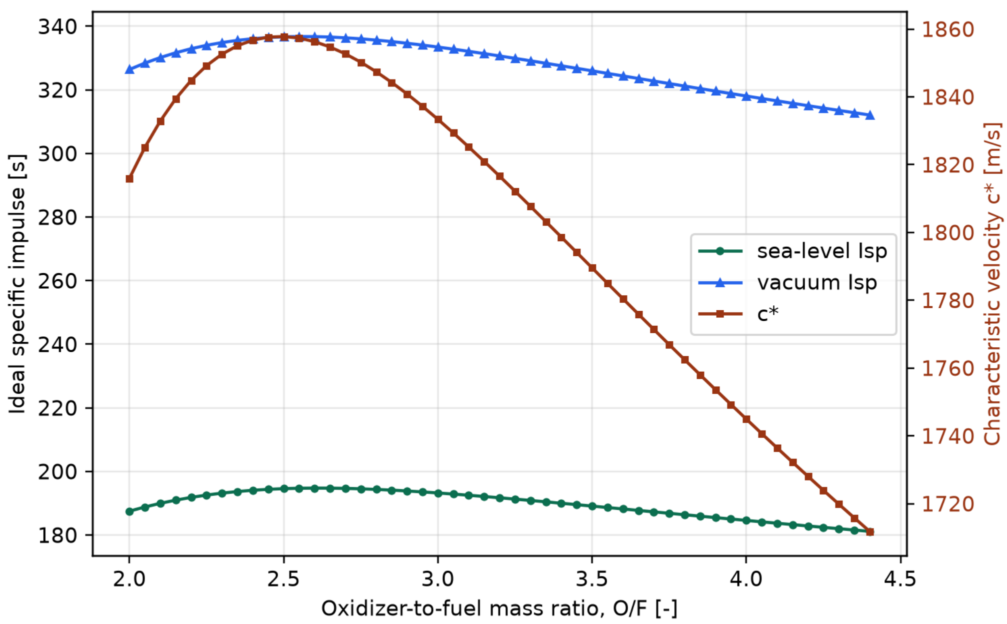

Figure 2: Ideal sea-level and vacuum specific impulse with characteristic velocity for a fixed nozzle area ratio of 15.

The behavior of γ and molecular weight is shown in Figure 3. The molecular weight increases with oxidizer content because the product mixture moves from H_2/CO-rich toward $H_2O/CO_2/O_2$-rich composition. The heat capacity ratio remains near 1.20-1.22 in the hottest part of the sweep, which is typical for high-temperature combustion products.

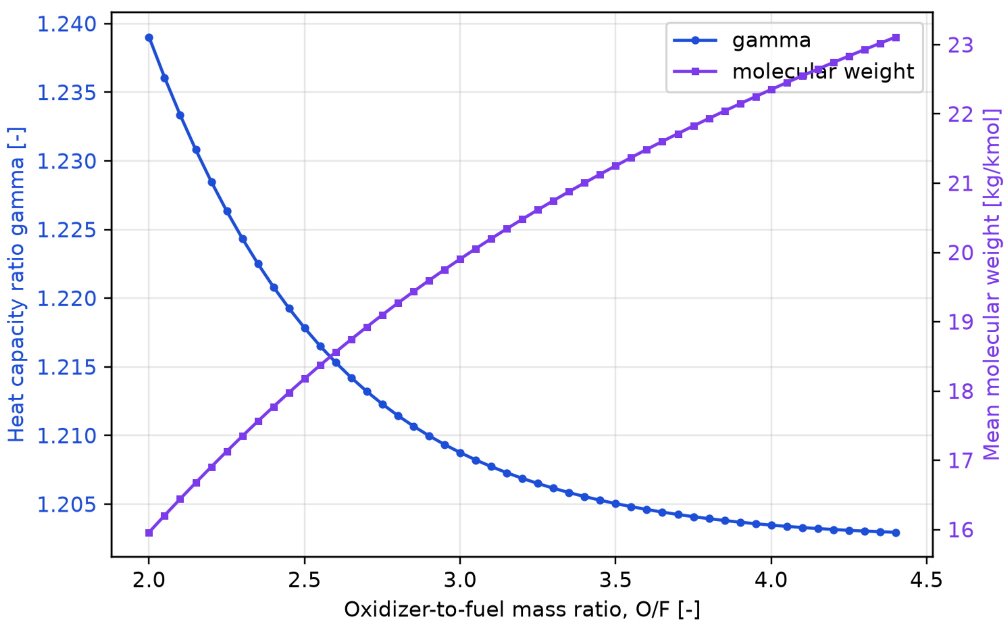

Figure 3: Equilibrium heat capacity ratio and mean molecular weight.

Figure 4 explains the fuel-rich performance optimum. At low O/F the products contain large mole fractions of H_2 and CO. Moving toward stoichiometric and oxidizer-rich operation increases H_2O, CO_2 and O_2. The result is a hotter but heavier gas mixture.

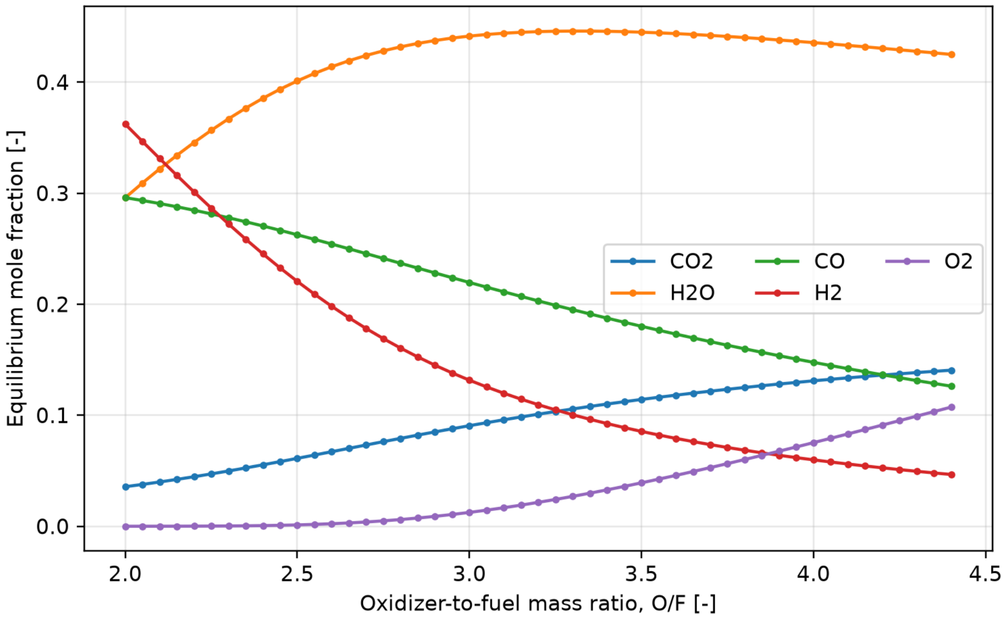

Figure 4: Major product mole fractions at chamber equilibrium.

Selected numerical results are summarized in Table 1. The stoichiometric region has the highest temperature, but not the highest ideal vacuum performance in this simplified model.

Table 1: Selected results for the 20 bar O/F sweep.

Case	O/F	T_c [K]	$\overline{M}$ [kg/kmol]	c^* [m/s]	$I_{sp,vac}$ [s]
Best $I_{sp,vac}$	2.55	3237.7	18.37	1857.3	336.8
Best T_c	3.70	3455.9	21.72	1771.4	322.8
Rounded stoichiometric	4.00	3450.2	22.35	1745.0	318.1
Best c^*	2.50	3207.6	18.18	1857.8	336.7

5.2 Pressure Sweep

The pressure sweep in Figure 5 was run at O/F = 2.55, the mixture ratio that maximized vacuum specific impulse in the baseline sweep. Increasing chamber pressure from 20 to 100 bar raises the equilibrium temperature from 3237.7 K to 3368.6 K. The vacuum specific impulse increases only modestly, from 336.8 s to 342.5 s. In contrast, sea-level specific impulse increases from 194.7 s to 313.6 s because the fixed area-ratio nozzle becomes less overexpanded as p_c/p_a rises.

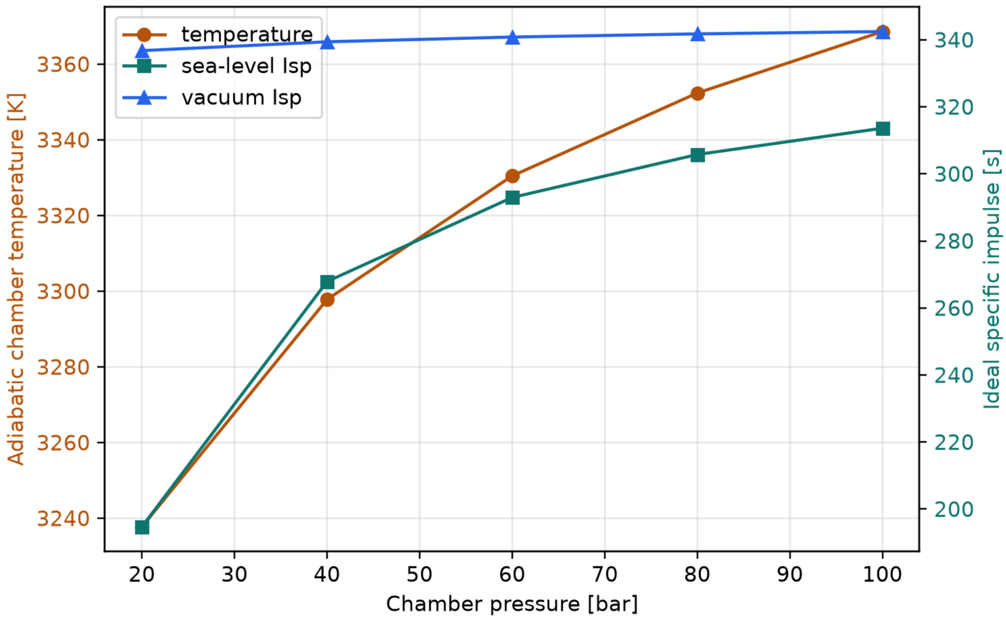

Figure 5: Influence of chamber pressure at O/F = 2.55.

Figure 6 shows the transport-property trend. Dynamic viscosity increases with temperature over most of the sweep, while the Prandtl number varies weakly compared with the composition and temperature changes. These quantities are relevant for later thermal or cooling-channel calculations, although no wall heat-transfer model is included in the present project.

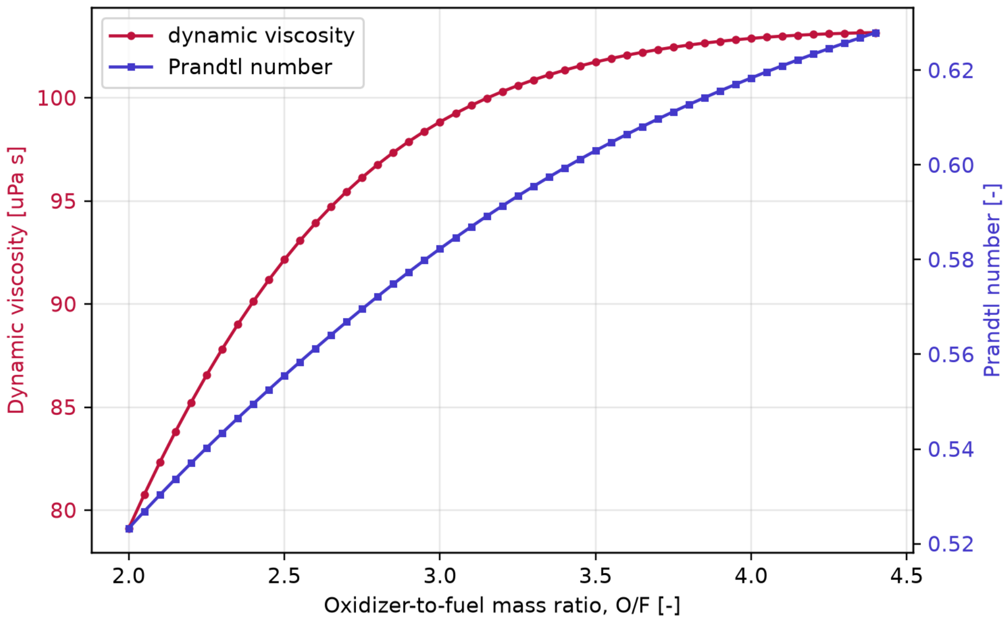

Figure 6: Dynamic viscosity and Prandtl number of the equilibrium products.

6 Limitations

The most important limitations are:

- the reactants are treated as ideal gases at 300 K, not as cryogenic liquids with realistic injection enthalpy;
- only gas-phase GRI-Mech species are considered, so condensed carbon is not modeled in very fuel-rich mixtures;
- the chamber is perfectly mixed and adiabatic;
- the nozzle model freezes γ, R and composition at chamber values instead of solving a full equilibrium or finite-rate expansion;
- no efficiencies, pressure losses, boundary-layer losses or heat losses are included.

These assumptions make the model suitable for trend analysis and coursework, but not for final engine sizing.

7 Conclusions

A complete Python/Cantera workflow for preliminary LOX/LCH_4 rocket combustion analysis was created. The code automatically sweeps mixture ratio and chamber pressure, computes equilibrium thermodynamic and transport properties, estimates ideal nozzle performance and generates figures for the report.

For the baseline chamber pressure of 20 bar, the maximum chamber temperature is 3455.9 K at O/F = 3.70. The highest ideal vacuum specific impulse in the simplified frozen-gamma nozzle model is 336.8 s at O/F = 2.55. The difference between these optima is an important engineering result: maximum temperature is not equivalent to maximum performance, because molecular weight and expansion properties also matter.

The pressure sweep shows that vacuum performance is only weakly affected by pressure over the tested range, while sea-level performance strongly improves with increasing chamber pressure for a fixed nozzle area ratio. This is mainly an expansion-matching effect rather than a large chemical-energy change.

Future work should add equilibrium nozzle expansion, realistic cryogenic reactant enthalpies, comparison with NASA CEA and a simple regenerative cooling or heat-flux model. Those extensions would make the tool more representative of real rocket engine design while keeping the present reproducible Python structure.

References

[1] T. Neill, D. Judd, E. Veith, and D. Rousar, "Practical uses of liquid methane in rocket engine applications," *Acta Astronautica*, vol. 65, no. 5–6, pp. 696–705, 2009.

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[3] Cantera Developers, "Cantera: An object-oriented software toolkit for chemical kinetics, thermodynamics, and transport processes," 2026. Accessed: 2026-06-14.

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Repository

Click to access the GitHub repository.